Tables, Color, and Figures in LaTeX

Contents

1 Creating a Basic Table

Definition. The `tabular` environment creates a table. The column format tells LaTeX how many columns there are and how to align them. The letter `l` means left, `c` means centered, and `r` means right.

Source code:

```
\begin{tabular}{|l|c|r|}
\hline
Name & Subject & Marks \\
\hline
Asha & Maths & 95 \\
Ravi & Science & 88 \\
\hline
\end{tabular}
```

Expected result:

Name	Subject	Marks
Asha	Maths	95
Ravi	Science	88

Commands. Use `&` to separate columns, `\\` to start a new row, and `\hline` to draw a horizontal line.

Practice. Create a table with three columns: item, quantity, and price.

2 Improving Tables with Booktabs

Definition. The `booktabs` package provides cleaner horizontal rules for professional-looking tables. Its main commands are `\toprule`, `\midrule`, and `\bottomrule`.

Source code:

```
\begin{tabular}{lcr}
\toprule
Name & Level & Score \\
\midrule
Mina & Beginner & 82 \\
```

```
Omar & Advanced & 96 \\
\bottomrule
\end{tabular}
```

Expected result:

Name	Level	Score
Mina	Beginner	82
Omar	Advanced	96

Commands. Add `\usepackage{booktabs}` in the preamble. Prefer these rules instead of drawing a box around every cell.

Practice. Make a booktabs table comparing three programming languages.

3 Adding Color to Text

Definition. The `xcolor` package allows text and other elements to be colored. The command `\textcolor` applies a color to a selected piece of text.

Source code:

```
\textcolor{red}{This text is red.}

\textcolor{blue}{This text is blue.}

\textcolor{teal}{This text is teal.}
```

Expected result:

This text is red.
This text is blue.
This text is teal.

Commands. Load the package with `\usepackage{xcolor}`. The syntax is `\textcolor{color}{text}`. Many standard color names are available.

Practice. Write one sentence in green and another in purple.

4 Highlighting Table Rows and Cells

Definition. With `xcolor`'s table support, you can shade an individual cell or an entire row. This can make headings and important values easier to notice.

Source code:

```
\begin{tabular}{|l|c|}
\hline
\rowcolor{lightgray}
Product & Stock \\
\hline
Notebook & 24 \\
\cellcolor{yellow}Pen & 50 \\
\hline
\end{tabular}
```

Expected result:

Product	Stock
Notebook	24
Pen	50

Commands. Use `\rowcolor{lightgray}` immediately after the row begins to shade a row. Use `\cellcolor{yellow}` inside a cell to shade only that cell.

Practice. Shade the heading row of your own table and highlight one important cell.

5 Including an Image

Definition. The `graphicx` package provides the `\includegraphics` command for inserting image files. Keep the image file in the same folder as your `.tex` file, or provide its relative path.

Source code:

```
\usepackage{graphicx}

\begin{figure}
\centering
\includegraphics[width=0.6\textwidth]{example-image}
\caption{An example image}
\label{fig:example}
\end{figure}
```

Expected result: A centered image appears at 60% of the text width, with the caption "An example image" below it. The sample filename `example-image` is supplied by many LaTeX installations; replace it with your own image filename, such as `photo.png`.

Commands. The `width` option controls image size. The `figure` environment creates a floating figure. Use `\centering` to center it, `\caption` to add a caption, and `\label` to give it a reference name.

Practice. Insert an image named `myphoto.png` and give it a meaningful caption.

6 Figure Placement and Resizing

Definition. Figures are floats, so LaTeX may move them to improve page layout. Placement options suggest where a figure should appear: `h` means here, `t` top, `b` bottom, and `p` a page of floats. The exclamation mark asks LaTeX to be less restrictive.

Source code:

```
\begin{figure}[htbp]
\centering
\includegraphics[width=0.45\textwidth]{example-image}
\caption{A resized figure}
\label{fig:resized}
\end{figure}
```

Expected result: The image is displayed at 45% of the text width, with a caption. LaTeX chooses a suitable location, preferring here, top, bottom, or a float page.

Commands. Change the width to `0.8\textwidth` for a larger image. Avoid forcing every figure into an exact position unless there is a good reason.

Practice. Add a second figure and set its width to 30% of the text width.

7 Referencing Figures and Tables

Definition. A label lets you refer to a numbered figure or table elsewhere in the document. Use `\ref` to print its number. LaTeX may need to be compiled twice to update references.

Source code:

```
Figure~\ref{fig:example} shows the sample image.

\begin{figure}
\centering
\includegraphics[width=0.4\textwidth]{example-image}
\caption{Sample image for reference}
\label{fig:example}
\end{figure}
```

Expected result: The sentence refers to the figure number automatically, for example, "Figure 1 shows the sample image."

Commands. Put `\label` after `\caption` so the label records the correct figure number. The same approach works for tables.

Practice. Add a table caption and label, then refer to it in a sentence using `\ref`.

8 A Complete Mini Example

The following example combines a colored heading, a table, and a figure. The TikZ drawing is used here so the example compiles without needing an external image file.

Source code:

```
\textcolor{blue}{\Large My Project Summary}

\begin{tabular}{lr}
\toprule
Task & Hours \\
\midrule
Planning & 3 \\
Writing & 5 \\
Review & 2 \\
\bottomrule
\end{tabular}

\begin{figure}[H]
\centering
\begin{tikzpicture}
\draw[fill=blue!20] (0,0) rectangle (3,1.5);
\draw[thick] (0,0) -- (3,1.5);
```

```
\draw[thick] (0,1.5) -- (3,0);
\end{tikzpicture}
\caption{A simple project illustration}
\end{figure}
```

Expected result:

My Project Summary

Task	Hours
Planning	3
Writing	5
Review	2

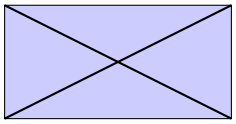


Figure: A simple project illustration

Practice. Create a one-page report containing a colored title, a table with at least three rows, and one labeled figure.